BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

PHYSICS | SSC-I (9th Grade) | Units 1-6 and 9 | Marking Report

Exam: exam-10-ch1-6-9 (Version 1) **Candidate:** Seemab **Date:** 16 July 2026 **Total Marks:** 65

Independently re-verified: marked page by page against the answer key, then every Section B and Section C calculation was re-derived from scratch (not just letter-matched to the key) – acceleration, momentum, moments, plumb-line method, three layered-pressure numericals, impulse, and energy numericals all recomputed independently and match Seemab's working exactly. Section A checked letter by letter against the key (11/12 confirmed, only Q7 disagrees). Both passes agree on **64/65 (98.5%)**.

64 / 65 98.5% A+

Section A: 11/12 | Section B: 33/33 | Section C: 20/20 | Only one mark lost on the entire paper

SECTION A: MCQS	SECTION B: SHORT	SECTION C: LONG
11/12	**33/33**	**20/20**
91.7% (1 miss – Q7)	100% (perfect)	100% (perfect, all 4)

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (abbreviated)	Seemab	Correct	Result
1	Instrument for timing an event to 0.1 s precision	A	A (Digital stopwatch)	CORRECT
2	Vernier caliper least count (0.5 mm scale, 25 divisions)	B	B (0.02 mm)	CORRECT
3	Motorboat acceleration, 8→26 m/s in 6 s	C	C (3 m/s²)	CORRECT
4	Time to reach max height, u = 25 m/s	D	D (2.5 s)	CORRECT
5	Trolley acceleration, F = 32 N, m = 8 kg	A	A (4 m/s²)	CORRECT
6	Forces in FBD of a book resting on a table	C	C (Weight + Normal force)	CORRECT
7	Centre of gravity of an L-shaped (bent) lamina	C	D (May lie outside the lamina)	INCORRECT
8	Torque, F = 25 N at 0.3 m	A	A (7.5 N·m)	CORRECT
9	Pressure of 0.76 m mercury column	B	B (1.03×10^5 Pa)	CORRECT
10	Power of pump, 3000 J in 20 s	C	C (150 W)	CORRECT
11	Energy resource from Earth's crust / radioactive decay	A	A (Geothermal)	CORRECT
12	Branch of physics studying blood flow / respiration	D	D (Biophysics)	CORRECT

MCQ score: 11/12. Only Q7 is wrong. Notably, Seemab marked her own answer sheet with a self-check **X** on Q7 (all 11 others got a ✓) – she flagged her own doubt on this one at the time but did not change the answer. This is Physics' **first-ever MCQ miss**: the subject's all-time perfect 84/84 (100%) record now becomes 95/96 (98.96% ≈ 99.0%).

Section B — Short Answer Questions (11 × 3 = 33 Marks)

All 11 parts attempted, every sub-part completed – no stopping-early this time. OR variant chosen for 2(i), 2(viii), 2(ix).

Q.2(i): Measuring Cylinder Method for an Irregular Solid *Chose: OR alternative* **3/3**

Correct method described (partly-filled cylinder, solid fully submerged, volume = displaced water). Initial reading 24 mL, final reading 63 mL → volume = 63 − 24 = **39 cm³**. Matches the key exactly.

Q.2(ii): Scientific Notation & SI Prefixes *Chose: Main* **3/3**

(a) 0.000085 m = 8.5×10^{-5} m, correctly re-expressed with a prefix as 85×10^{-6} m = 85 µm. (b) 52,000,000 s = 5.2×10^7 s. Both correct.

Q.2(iii): Uniform Acceleration Numerical *Chose: Main* **3/3**

(a) a = (26−6)/8 = **2.5 m/s²**. (b) s = ut + ½at² = 6(8) + ½(2.5)(64) = 48 + 80 = **128 m**. A crossed-out false start is visible above the working, but the final boxed answer is correct and matches the key.

Q.2(iv): Distance & Displacement of a Jogger *Chose: Main* **3/3**

(a) Total distance = 500 + 200 = **700 m**. (b) Displacement = 500 − 200 = **300 m**. Both correct, with distance/displacement correctly distinguished as scalar vs. net position.

Q.2(v): Newton's Second Law – Trolley Acceleration *Chose: Main* **3/3**

Law stated correctly (acceleration directly proportional to net force, inversely proportional to mass). (a) a = 42/7 = **6 m/s²**. (b) Same force, m = 3.5 kg: a = 42/3.5 = **12 m/s²**. Both correct.

Q.2(vi): Momentum Change & Average Force (Hockey Strike) *Chose: Main* **3/3**

(a) Δp = m($v_f - v_i$) = 0.16 × 30 = **4.8 kg·m/s**. (b) F = Δp/Δt = 4.8/0.02 = **240 N**. Both correct.

Q.2(vii): Principle of Moments (See-saw) *Chose: Main* **3/3**

Clockwise moment = anticlockwise moment stated. $300 \times 0.8 = 400 \times d \rightarrow d = 240/400 =$ **0.6 m**. Correct, full working shown.

Q.2(viii): Finding the Centre of Gravity of an Irregular Lamina *Chose: OR alternative* **3/3**

Correct definition of CG, then the full plumb-line method: hole near the edge, freely suspended, plumb line hung and traced, repeated from ≥2 more points, intersection = CG. All steps present and correct.

Q.2(ix): Friction & Heat – Rubbing Hands *Chose: OR alternative* **3/3**

Correctly explains that rough skin surfaces resist relative motion (friction), work done against friction is dissipated as heat, and correctly names the energy conversion (mechanical → thermal). Full marks.

Q.2(x): Liquid Pressure – Kerosene Tank *Chose: Main* **3/3**

$P = \rho gh = 800 \times 10 \times 4 =$ **32,000 Pa**. Correct (the scientific-notation restatement "3.2×10^4" trails off at the very edge of the photo frame, but the primary numeric answer 32,000 Pa is fully legible and correct, so no mark is affected).

Q.2(xi): Pressure Definition & Tractor Numerical *Chose: Main* **3/3**

$P = F/A$ defined correctly, SI unit pascal (1 Pa = 1 N/m²) stated. $F = mg = 2500 \times 10 = 25{,}000$ N. $P = 25{,}000/0.5 =$ **50,000 Pa**. Correct.

Section B total: 33 / 33 (100%) – every one of the 11 questions fully correct

Section C — Long Answer Questions (4 × 5 = 20 Marks)

All 4 long questions attempted, every sub-part completed. OR variant chosen for 3(i), 3(ii), 3(iv).

Q.3(i): Three Equations of Motion & Stone Dropped from a Tower *Chose: OR alternative* **5/5**

(a) All three equations of motion listed with a symbols table (v_i, v_f, a, t, s all defined). (b) u = 0, g = 10, t = 5 s: $h = 0(5) + ½(10)(25) =$ **125 m**; $v = 0 + 10(5) =$ **50 m/s**. Both correct, full marks. (Minor notation note, not marked down: the middle equation is written as "$s = vt + ½at^2$" using an unsubscripted v rather than v_i – the numeric work itself correctly substitutes the initial velocity, so nothing was lost.)

Q.3(ii): Impulse-Momentum Relation & Karate Strike *Chose: OR alternative* **5/5**

(a) Impulse defined (product of net force and time) and correctly derived from $F = ma$ through $F\Delta t = \Delta p$. (b) m = 1.2 kg, v_i = 12 m/s, v_f = 0, Δt = 0.008 s: (i) Impulse = $1.2(0-12) = -14.4$ kg·m/s, magnitude **14.4 N·s**. (ii) $F = 14.4/0.008 =$ **1800 N**. Both correct.

Q.3(iii): Derivation of P = ρgh & Layered Oil/Water Tank *Chose: Main* **5/5**

(a) Full derivation from first principles: $V = Ah$, $m = \rho Ah$, $F = mg = \rho Ahg$, $P = F/A = \rho gh$. (b) (i) Oil layer alone: $800 \times 10 \times 0.5 =$ **4000 Pa**. (ii) Total at bottom: $4000 + (1000)(10)(2.5) = 4000 + 25{,}000 =$ **29,000 Pa**. Both correct, correctly distinguishing "oil alone" from "total including water beneath."

Q.3(iv): Renewable vs. Non-renewable Energy & Hydroelectric Turbine *Chose: OR alternative* **5/5**

(a) Correct comparison table (naturally replenished / cannot run out vs. limited, formed over millions of years, cannot be replaced), with valid examples (biomass; nuclear fuel). (b) PE lost per second = $mgh = 2000 \times 10 \times 15 =$ **300,000 J**. Power = 300,000/1 = **300,000 W = 300 kW**. Both correct.

Section C total: 20 / 20 (100%) – all four long questions fully correct

Complete Deduction Log (reconciles to headline score)

Question	Awarded	Lost	Reason
Section A, Q7 (MCQ)	0/1	−1	Centre of gravity of an L-shaped (bent) uniform lamina. Correct answer is D – "May lie outside the lamina, in empty space" (the textbook's Fig 4.7 explanation for a bent-arm CM/CG). Seemab chose C – "Cannot be determined for irregular shapes." She self-marked an X next to this one on her own answer sheet (all 11 other MCQs got a ✓), showing she doubted the answer while writing it but did not revise it.
Total deductions	**−1**		**65 − 1 = 64/65 confirmed. Every Section B and Section C question scored full marks – this is the only mark lost on the entire paper.**

Strengths in this attempt

- **Perfect Section B (33/33) and perfect Section C (20/20)** – only the second time both sections have been simultaneously flawless in Physics (after exam-03's clean 100%).
- Every sub-part of every question attempted and completed – the "stopping before finishing every sub-part" pattern tracked across her file did not appear anywhere on this paper.
- Full first-principles derivations correct: $P = \rho gh$ from a liquid column (3(iii)), and the impulse-momentum relation $F\Delta t = \Delta p$ from Newton's second law (3(ii)).
- Layered-liquid pressure numerical (oil over water) correctly separated "oil layer alone" from "total pressure at the bottom."
- Complete, correctly-sequenced plumb-line method for finding the CG of an irregular lamina.
- Second-highest Physics score to date (98.5%, behind only exam-09's 99.2%).

Areas to revise

- **Centre of gravity of bent/irregular shapes (the only mark lost):** for an L-shaped lamina, the CG can fall *outside* the actual material, in the empty space of the bend – it is not simply "impossible to determine." Revisit the textbook's bent-arm CM/CG figure; this is a clean, revisable concept gap, not a careless slip.
- **Tighten equation notation (no marks lost, but worth cleaning up):** in the general equations-of-motion list, she wrote "$s = vt + ½at^2$" using a bare v rather than v_i for initial velocity, while everywhere else she consistently used v_i/v_f. A stricter examiner could read the bare v as final velocity. Her actual numeric substitution was correct, so this is a habit to tidy rather than a concept problem.

Physics Exam Comparison

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	7 Apr 2026	63.5/65	97.7%	12/12	32.5/33	19/20	A+
exam-02-ch3-4-9	10 Apr 2026	62/65	95.4%	12/12	30.5/33	19.5/20	A+
exam-03-ch5	18 May 2026	65/65	100%	12/12	33/33	20/20	A+
exam-04-ch6	22 Jun 2026	55.5/65	85.4%	12/12	29/33	14.5/20	A
exam-07-ch1-6-9	25 Jun 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+

exam-08-ch1-6-9	2 Jul 2026	61/65	93.8%	12/12	30.5/33	18.5/20	A+
exam-09-ch1-6-9	7 Jul 2026	64.5/65	99.2%	12/12	32.5/33	20/20	A+
exam-10-ch1-6-9 (this)	**16 Jul 2026**	**64/65**	**98.5%**	**11/12**	**33/33**	**20/20**	**A+**

exam-10 (64/65, 98.5%) is Seemab's **second-highest Physics score to date**, with a flawless Section B and Section C – the MCQ slip on Q7 is the only thing separating it from a second 100%. This also ends Physics' flawless all-time MCQ record (was 84/84, 100%; now 95/96, 98.96%). Physics' subject average rises to **95.0% (494.0/520, 8 exams)**, overtaking Mathematics (94.5%) to become Seemab's strongest subject outright.